

Improving the Robustness of Reservoir Operations with Stochastic Dynamic Programming

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Abstract: Reservoir operations should consider both adaptiveness and robustness to deal with two of the main characteristics of climate change: nonstationarity and deep uncertainty. In particular, robust operational strategies are distinguished from risk-neutral expected value optimization in the sense that they should be satisfactory over a wider range of uncertainty and improve the ability of a reservoir system to adapt to climate change. In this study, a new framework named robust stochastic dynamic programming (RSDP) is proposed that couples robust optimization (RO) with the formulations of objective function or constraints used in stochastic dynamic programming (SDP). Two main approaches of RO, namely feasibility robustness and solution robustness, are both considered in the optimization algorithm. Consequently, this study uses the Boryeong multipurpose dam to evaluate three SDP framings: conventional-SDP (CSDP), RSDP-feasibility robustness (RSDP-F), and RSDP-solution robustness (RSDP-S). These three SDP formulations were used to derive optimal monthly release rules for the Boryeong Dam, and their relative performances were evaluated using simulations of a broader range of inflow scenarios. The simulation-based re-evaluations of the resulting reservoir operational policies were quantified using a wide range of metrics that include reliability, resiliency, and vulnerability, as well as regret-based robustness metrics. The results of this study suggest that the RSDP-S model not only increases the range of possible solutions, but also yields more desirable operation outcomes under extreme climate conditions with respect to both traditional and robustness metrics. DOI: [10.1061/\(ASCE\)WR.1943-5452.0001381](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001381). © 2021 American Society of Civil Engineers.

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Introduction

Securing stable freshwater sources through water resource infrastructure, such as reservoirs, levees, and river regulators, has been an important task for humanity since the beginning of civilization. Nevertheless, recently, increases in the frequency of extreme floods and droughts (Loucks 2000; Cosgrove and Loucks 2015), population growth and urbanization, and climate change (Salazar et al. 2016) have caused traditional planning and management frameworks for water resource infrastructure to evolve to be able to more effectively deal with these compounding challenges (Bouwer 2000; Li et al. 2010; Poff et al. 2016; Brown et al. 2015; Quinn et al. 2018). Climate change is difficult to address due to two of its main characteristics: nonstationarity (Milly et al. 2008) and deep (or Knightian) uncertainty (Lempert 2014; Dittrich et al. 2016). In the field of water resources, nonstationarity commonly refers to changes in the statistical properties of random variables with shifts in time (Razavi et al. 2015), while deep uncertainty accounts for a situation in which analysts do not know or cannot agree on probabilistic distributions of key random variables and parameters

(Lempert 2003; Ben-Haim 2006; Hallegatte et al. 2012; Brown et al. 2020). Numerous studies conducted over the past two decades, which have been focused on planning and managing water resources systems during climate change, also stress the importance of considering nonstationarity and deep uncertainty in adaptation strategies (Haasnoot et al. 2013; Steinschneider and Brown 2012; Herman et al. 2015, 2020; Moallemi et al. 2020).

Given challenging hydroclimatic variabilities and uncertain demands impacting real-world water resources systems (Johnson et al. 1991; Trindade et al. 2017; Fletcher et al. 2019; Woodward et al. 2014), scholars have long agreed on the necessity of seeking robust decisions that are “insensitive to errors, random or otherwise, in the estimates of those parameters affecting design choice” (Matalas and Fiering 1977; Herman et al. 2015; Giuliani and Castelletti 2016) when compared with traditional least-cost optimization approaches. Given that there can be diverse concerns across the economic, engineering, and socioecological objectives of water systems (Botte and Schöbel 2019), recent studies have highlighted the importance of avoiding myopic definitions of robustness that fail to account for diverse stakeholder interests and needs (Kasprzyk et al. 2013; Herman et al. 2014; McPhail et al. 2018; Hadjimichael et al. 2020). Early conceptions of robustness in the optimization literature by Mulvey et al. (1995) and Watkins and McKinney (1997) sought to move beyond traditional deterministic least-cost optimization by defining robust optimization (RO), which seeks to better address risk aversion by changing the formulations of mathematical programming models. Recently, Ray et al. (2014) applied RO to a city’s water infrastructure planning example with two conflicting objectives by including both solution robustness (penalizing variance of the objective) and feasibility robustness terms (penalizing violation of the constraints) into the objective function of the model and demonstrated a trade-off relationship regarding stakeholder preferences. Ray et al. (2014) illustrated the necessity of adopting robust decisions despite the unlikelihood of a catastrophe

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with their experiment. Roach et al. (2016) compared the performance of RO with a different set of robust decision theory, info-gap (IG) (Ben-Haim 2006), and concluded that solutions derived using RO cover a wider range of varying target robustness levels, whereas solutions produced by IG yielded dominant solutions under a limited range. However, this study took only the financial aspects of the strategy into consideration when applying RO, which may result in a different set of solutions to be implemented in a real-world system. Yet, to date no study has coupled RO to a real-world reservoir operation problem in which the primary concepts of RO can be incorporated into the problem formulation and optimization processes.

Over the last 50 years, reservoir operations have been one of the most active areas of research in the water systems literature (Yeh 1985; Labadie 2004; Quinn et al. 2019; Raso et al. 2019; Raso et al. 2020). Recent studies are focused on two main topics: forecast-informed reservoir operations (Tejada-Guibert et al. 1995; Kim and Palmer 1997; Zhao et al. 2011; Anghileri et al. 2016; Turner et al. 2017) and advancing optimization techniques that can broaden the array of decision relevant concerns that can be incorporated into the design of operations (Castelletti et al. 2010, 2013; Giuliani and Castelletti 2016; Quinn et al. 2019).

This study contributes a robust stochastic dynamic programming (RSDP) approach in which mathematical optimization concepts from RO are integrated into an SDP framework to identify reservoir control strategies that balance optimality and robustness. A conventional SDP (CSDP) baseline model is compared with two alternative robust SDP (RSDP) models that each emphasize two main concepts of robustness in mathematical programming as proposed by Mulvey et al. (1995): (1) solution robustness (nearness to optimality across all scenarios), and (2) feasibility robustness (nearness to feasibility across all scenarios). For more details on solution robustness and feasibility robustness, see the next section of this paper.

The primary contribution of this paper is an extension of SDP to identify both optimal and robust reservoir operation rules to better confront the nonstationarity and deep uncertainty associated with climate change. This study proposes an integrated framework of SDP and RO for deriving optimal reservoir operating policies that are then further evaluated with regret-based robustness evaluation metrics under a wide range of inflow scenarios. This work is an advancement of studies conducted by Ray et al. (2014) and Pan et al. (2015) in which only traditional evaluation metrics such as sustainability, reliability, and vulnerability were considered for evaluation, or in which only a limited concept of robustness was considered.

The remainder of this paper is organized as follows. In the next section, the key components and concepts of the main algorithms used in this study are introduced: SDP, RO, and the proposed RSDP. Then, the Boryeong Dam test case, as well as the specific mathematical formulations of its operations are described. Next, the results of optimization, sample operation under specific drought incidents, and metric evaluations are presented, highlighting the potential benefits as well as challenges that emerge with RSDP operational alternatives. Finally, this study concludes with reviewing key findings, possible shortcomings of this research, and suggested future work.

Methodology

Stochastic Dynamic Programming

SDP, based on the concept of Bellman's principle of optimality, is one of the most popular optimization algorithms used, by far, for the derivation of optimal reservoir operation policies (Stedinger et al. 1984; Kim et al. 2007; Haguma et al. 2018). Complex reservoir

operational design problems are nonlinear, stochastic, and multiobjective (Giuliani and Castelletti 2016), which require optimization algorithms that are capable of considering diverse types of objectives and uncertainties. Since SDP models can incorporate these complexities through discretization of the state space and use of sampling schemes based on statistical methods, they have been continuously used for solving reservoir operation problems. Recently, based on Butcher (1971), who first introduced the use of the discrete, lag-one Markov process to describe uncertainty in future streamflow, advanced variations, such as SDP models with multiple uncertainties (Shokri et al. 2013), stochastic dual dynamic programming (SDDP) (Gjelsvik et al. 2010; Raso et al. 2017) has been adopted for analyzing complex water resource systems. As SDP has been continuously used throughout the history of reservoir operations as previously cited, this paper will only briefly describe the main components of the algorithm that are used in the study, instead of explaining all fundamental concepts behind SDP. The reader is referred to review papers by Yakowitz (1982), Yeh (1985), Kim and Palmer (1997), Labadie (2004), and Castelletti et al. (2008) for more detailed explanations regarding SDP for reservoir operation and related applications.

In general, a steady-state monthly release policy, which is the result of optimization, is determined by solving the following backward recursive equation in a classical SDP model (Tejada-Guibert et al. 1995):

$$f_{opt}^n(S_t, H_t) = E_{Q_t|H_t} \left\{ \max_{R_t^*} [B_t(S_t, Q_t, R_t) + E_{H_{t+1}|H_t, Q_t} [f_{opt}^{n-1}(S_{t+1}, Q_t)]] \right\} \quad (1)$$

where t = month index; $t = 1, \dots, T$; T = final time step; n = number of time periods remaining until the end of the planning horizon; S_t = storage at the beginning of the month t ; R_t = actual release during month t ; Q_t = inflow during month t ; H_t = hydrologic state variable for month t ; $B_t(\cdot)$ = value of benefit function during month t ; f_{opt}^n = expected value of benefit function from current period to the end of the planning horizon; and $E_{Q_t|H_t}$ = conditional expectation operation for inflow of Q_t during month t , given H_t .

The decision variables of Eq. (1) are subject to the following feasibility constraint:

$$\max(R_{\min}, S_t + Q_t - S_{\max}) \leq R_t \leq \min(R_{\max}, S_t + Q_t - S_{\min}) \quad (2)$$

where $R_{\min}$, $R_{\max}$, $S_{\max}$, and $S_{\min}$ denote the minimum release, maximum release, maximum storage, and minimum storage of the reservoir, respectively.

Tejada-Guibert et al. (1995) suggested that related stakeholders should choose the hydrologic state variable in SDP's recursive equation, H_t , mainly according to the characteristics of the system to be optimized. Among various choices, the most common choice for the hydrologic state variable was determined to be the current time step's inflow Q_t and the previous time step's inflow Q_{t-1} . In this study, the current month's inflow, Q_t , is selected and included as a hydrologic state variable in the recursive equation, H_t in Eq. (1). Also, the inclusion of Q_t as a hydrologic state variable makes it possible for the correlation between two consecutive months to be included inside the recursive equation (Kim and Palmer 1997). Transition probability of inflow, $\lambda(Q_{t+1} = I|Q_t)$, which reflects the potential of the system to generate inflows with respect to the previous state (Haguma et al. 2018), is calculated according to the historical average monthly inflow records and included in the optimization process. Therefore, Eq. (1) can be modified as the following equation, which is the main recursive equation

used in this paper in all models for the derivation of optimal monthly release rules:

$$f_{opt}^n(S_t, Q_t) = \max_{R_t^*} \left[B_t(S_t, Q_t, R_t) + \sum_{Q_{t+1}} (\lambda(Q_{t+1} = l | Q_t) \times f_{opt}^{n-1}(S_{t+1}, Q_{t+1} = l)) \right] \quad (3)$$

Robust Optimization

RO is a concept introduced by Mulvey et al. (1995) in which the optimization algorithm aims to integrate goal programming formulations with a scenario-based description of the problem data. RO has been applied in combination with various other methods in the field of water resources, including the use of stochastic optimization techniques (Mulvey et al. 1995; Ray et al. 2014), the use of the fuzzy approach (Li et al. 2006; Jia and Culver 2006; Xie et al. 2017), and to deal with uncertainties in scenarios (Xu et al. 2009; Kang and Lansey 2013).

For an illustration of RO, consider a typical stochastic programming model where a scenario set of uncertain variables is defined as $\Omega = \{s_1, \dots, s_s\}$. The objective function and corresponding constraints are formulated as

$$\text{Minimize}_{x,y} \quad z_s = c^T y + f(x_s) \quad (4)$$

subject to

$$A y = b \quad (5a)$$

$$g(x_s) + B_s y = d_s, \quad s \in \Omega \quad (5b)$$

$$x_s, y \geq 0, \quad s \in \Omega \quad (5c)$$

where z_s = value of the objective function in scenario S ; x_s = vector of control variables in scenario S ; y = vector of decision variables; c = vector of cost coefficients; $f(x_s)$ = operation cost of the system in scenario S ; A = matrix for structural constraint of model; b = corresponding right-hand-side vector of structural constraint; $g(x_s)$ = function of response of the system to the value of control variables in scenario S ; B_s = matrix for effects of structural decisions on the system performance in scenario S ; and d_s = corresponding RHS vector in scenario S .

The typical stochastic programming model described above has been used in the past and yielded optimal solutions under circumstances, in which the status quo state of the world was guaranteed. However, the constraint in the control variables in Eq. (5b) can strictly limit and consequently discard possible potential solutions that may be more robust than prespecified optimal solutions. To overcome this limitation of typical stochastic programming formulation, RO allows for potential violations of Eq. (5b) by transforming it into a soft constraint, which is flexible over a scenario set at a cost, $\rho(\alpha_1, \dots, \alpha_s)$. A new infeasibility variable set $\{\alpha_1, \dots, \alpha_s\}$ is introduced and included in the modified objective function, which acts as a penalty function that is proportional to the magnitude of the constraint violation, as in the following equations:

$$\text{Minimize}_{x,y} \quad \sigma(z_1, \dots, z_s) + \omega \times \rho(\alpha_1, \dots, \alpha_s) \quad (6)$$

subject to

$$A y = b \quad (7a)$$

$$g(x_s) + B_s y + z_s = d_s, \quad s \in \Omega \quad (7b)$$

$$x_s, y \geq 0, \quad s \in \Omega \quad (7c)$$

where $\sigma(z_1, \dots, z_s)$ = aggregate objective function for all scenarios; $\rho(\alpha_1, \dots, \alpha_s)$ = feasibility penalty function that penalizes the violation of the soft constraint in all scenarios; and ω = weight indicating the degree of acceptance or rejection of infeasibilities in the solution set (Watkins and McKinney 1997).

By transforming the objective function and constraints of a typical stochastic programming model into those of the RO model, additional features that could not be included in the past can now be implemented. Mulvey et al. (1995) and Watkins and McKinney (1997) suggested various mathematical formulations of the aggregate objective function, such as worst case, expected utility, and mean/variance models, so that the risk-averse nature of decision-makers in the field of water resources can be captured during the optimization processes. There are two broad classifications of robustness that can be incorporated into the RO model through objective functions. The first and more frequently applied is feasibility robustness in which the range of search for a potential solution is expanded by the use of soft constraints and corresponding penalty terms (Xu et al. 2009; Mortazavi-Naeini et al. 2015). Although less often applied, the second is solution robustness in which the variance of the initial objective, z_s , is penalized according to the distance from either its best value ($\max z_s$) or its most frequent value [$\arg_{z_s} \max p(z_s)$, where $p(\cdot)$ represents the probability of occurrence] through the use of higher moments in the objective function (Watkins and McKinney 1997; Kasprzyk et al. 2009; Ray et al. 2014). Consequently, this paper aims to contribute to the literature on water resources by introducing a novel methodology that modifies the objective function and the constraints of SDP through the inclusion of both feasibility and solution robustness, which is to be explained in detail in the following subsection.

Robust Stochastic Dynamic Programming

RSDP, the optimization algorithm proposed in this paper, couples the key components of RO into the objective function and constraints of SDP. Although redefining the key components of the objective function may be limited in specific optimization algorithms, it was possible by selecting SDP as this study's main tool mainly due to its wide applicability over various types of objective functions. Moreover, when it comes to real-world implementations of a theoretically optimized policy, Yeh (1985) and Wurbs (1993) argued the importance of considering the gap between the two. Thus, this study suggests a policy similar to real-world implementable policies through reflection of historical operation schemes.

Due to its nature, transforming the constraints of the typical SDP model given in Eq. (2) to a soft constraint, as suggested in the RO formulation in Eq. (7b), is feasible but involves several steps. For instance, consider an example of a reservoir for which the former maximum storage, $S_{\max}$ is enlarged to $S_{\max,ext}$ through the construction of additional infrastructure. Then, the feasibility constraint of SDP [Eq. (2)] is primarily transformed into the following:

$$\max(R_{\min}, S_t + Q_t - S_{\max,ext}) \leq R_t \leq \min(R_{\max}, S_t + Q_t - S_{\min}) \quad (8)$$

Consequently, the benefit function of the SDP model originally defined as $B_t(\cdot)$ can be modified to include a penalty function so that it can include a penalty term of either violations of soft constraints in case of feasibility robustness or variance of the objective across different scenarios for solution robustness. Because specific mathematical formulas for both feasibility and solution RSDP models may vary according to the system, specific formulas adopted in this paper will be demonstrated through the derivation

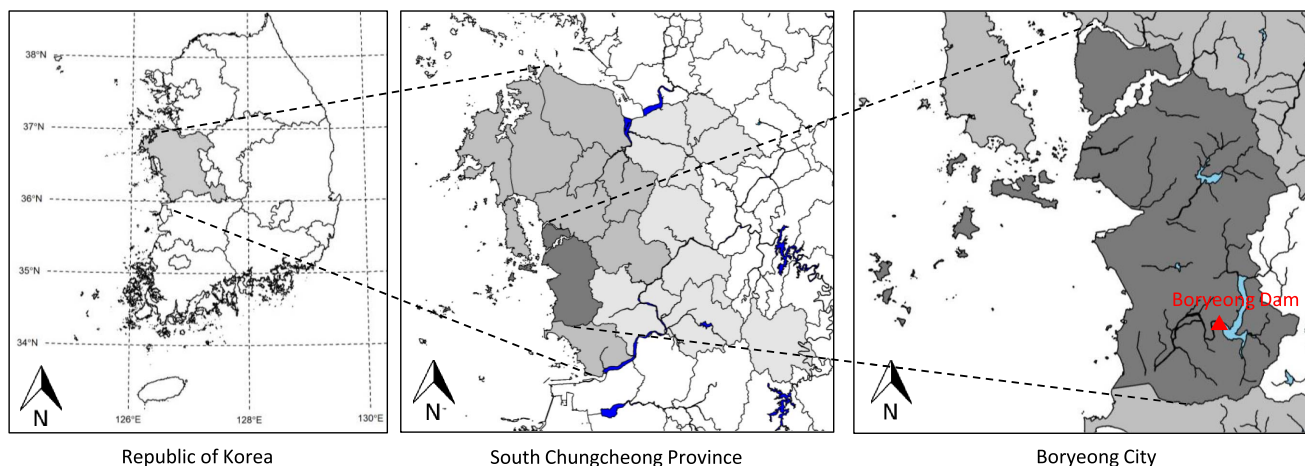


Fig. 1. Location of study area. Map of South Chungcheong Province (light gray) and the eight cities and counties (gray) that receive water from the Boryeong Dam (triangle) are displayed. City of Boryeong (dark gray) in which Boryeong Dam is included is distinguished from others.

of monthly release rules in a multipurpose dam in Korea in the following case study.

Case Study: Boryeong Multipurpose Dam

Study Area

The proposed RSDP is applied to derive monthly release rules for the Boryeong Dam, a multipurpose facility that supplies municipal and industrial (M&I) water to eight administrative cities and counties located on the west coast of the Korean peninsula (Fig. 1). The drainage area of the Boryeong Dam is relatively small, 163.6 km², and it receives an average of 1,245.0 mm of precipitation annually with annual runoff of 236.84×10^6 m³/year and dry season runoff of 64.65×10^6 m³/year. Furthermore, the capacity of the Boryeong Dam is 116.9×10^6 m³, and it is also responsible for supplying 106.6×10^6 m³ of water annually (Jung et al. 2016). In addition to concentrated precipitation during the 3-month flooding season (July–September), this narrow gap between the capacity of the reservoir and annual water supply leads to a high vulnerability of the dam to droughts if the reservoir is not operated properly during the 9-month drawdown period (October–June of the following year). Most of the water from the Boryeong Dam is used for M&I use (84%), whereas the ratio for other uses such as environmental (11%) and agricultural (5%) are relatively low (Jung et al. 2016) (Fig. 2), resulting in extremely risk-averse citizens who are highly sensitive to a shortage in usable water.

The acquirement of additional sources of freshwater through the construction of new infrastructure or an extension of the existing reservoir capacity is limited, especially close to the study area where stakeholders with different backgrounds compete with each other (Lee et al. 2019). Recently, when the cities and counties relying on the Boryeong Dam had to experience a severe multiyear drought during 2014–2019, various efforts were taken to mitigate the damage from the drought, such as the development of a water policy council (Kim et al. 2019a, b), construction of a conduit to act as an additional water source (Ihm et al. 2019), and improvements of the operating rules of the reservoir (Yoon et al. 2018). More practically, and related to the implemented policy, due to the national scale of the recent drought, Korea Water Resources Cooperation (KWRC) (Lee et al. 2019) has computed the benefits from making use of the emergency storage, which is the difference in reservoir

storage between the low water level and the dead storage level (Fig. 2) in the case of droughts. Detailed collaborative stakeholder elicitations (Kim et al. 2019b) informed the problem formulations explored in this study. Building on these stakeholder interactions, this paper aims to improve the efficacy of real-world reservoir operations by contributing optimization techniques that allow applications to better take advantage of recent innovations in robustness concepts.

An application of the proposed methodology is composed of three sequential steps: (1) optimization, (2) simulation, and (3) evaluation. Derivation of optimal monthly release rules according to each model, as well as the evaluation of the number of infeasible paths, is conducted in the optimization step. Then, the optimized rules are simulated under a wide range of inflow scenarios that are statistically generated to reflect the nonstationarity of climate change, thereby reflecting real-world policy that is actually being implemented. Ultimately, the simulated results are evaluated in the last step using various metrics. The general flow chart of the case study is shown in Fig. 3.

Optimization

Model and Objective Function Formulation

For comparative evaluation, three models were selected and the objective function and constraints were formulated corresponding to each model type: CSDP, robust SDP-feasibility robustness (RSDP-F), and robust SDP-solution robustness (RSDP-S). The models differ from each other mainly according to the terms that are included inside the objective function and the type of constraint that is implemented. Since the primary target of the reservoir in the study basin is to meet aggregate water demand for all users, the main objective term f_1 , included in all the models, is to minimize the deficit in the monthly release, R_t , with respect to the demand, D_t , across all time steps

$$f_1 = \max \left(\frac{D_t - R_t}{D_t}, 0 \right) \quad (9)$$

The second objective term, corresponding to the key concept of feasibility robustness, is introduced by advancing the research conducted by Lee et al. (2019), by modifying certain features so that the derived rules would be implementable to the application site. In this application, whereas the minimum storage for CSDP model is low water level (LWL), RSDP-F and RSDP-S models are allowed

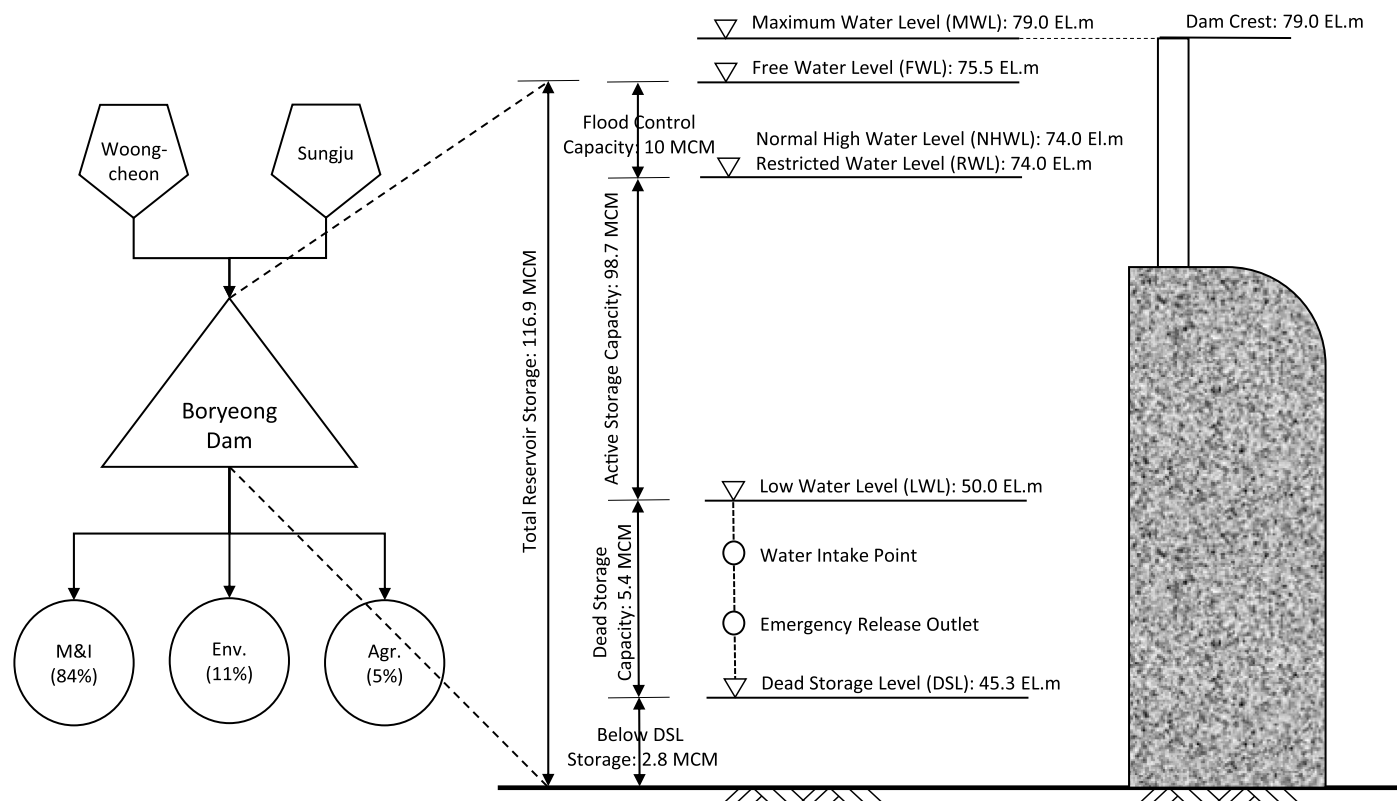


Fig. 2. Schematic diagram and water levels of Boryeong Dam. CSDP model may operate between FWL and LWL, while RSDP-F and RSDP-S models can be operated between FWL and DSL with the inclusion of a penalty terms which penalize the magnitude of violations of storage under LWL.

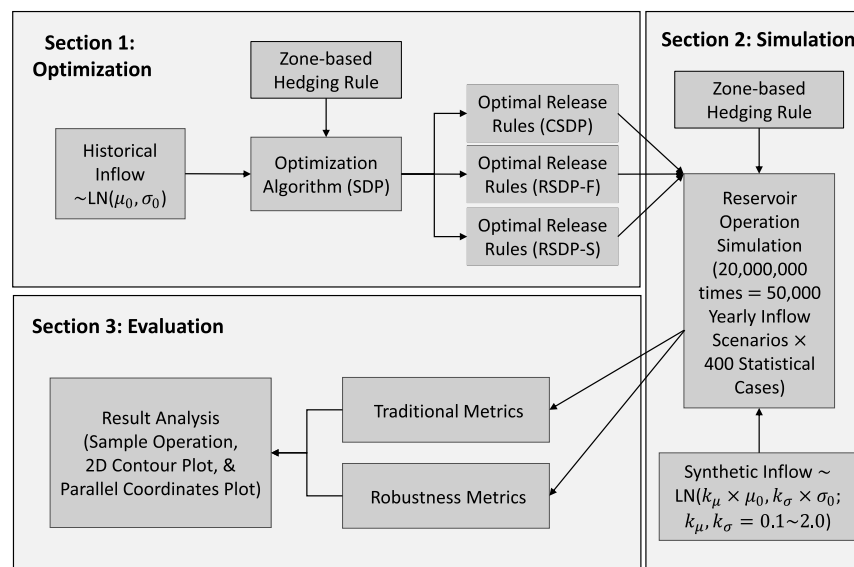


Fig. 3. Overall flow of the study. Case Study Section is organized as in steps in Fig. 3: (1) optimization; (2) simulation; and (3) evaluation.

to use the storage below LWL and above dead storage level (DSL) by transforming $S_{\min}$ into $S_{\min,R}$ (Fig. 2). By allowing violation of constraint through the use of a soft constraint, a corresponding penalty term is included in the objective function to measure the magnitude of the violation. Thus, the second term, which is included in RSDP-F and RSDP-S models, is defined as f_2

$$f_2 = \max\left(\frac{S_{\min} - S_t}{S_{\min}}, 0\right) \quad (10)$$

Due to this consideration, the former constraint given as Eq. (2), should be modified accordingly so that a fraction of all state variables in the RSDP models are able to pass through. As this

Table 1. Weights of objective terms in each optimization model

Model name	w_1	w_2	w_3	w_4
Conventional SDP (CSDP)	1×10^6	0	0	1
Robust SDP-feasibility robustness (RSDP-F)	1×10^6	1×10^6	0	1
Robust SDP-solution robustness (RSDP-S)	1×10^6	1×10^6	1×10^6	1

application allows for the usage of storage below LWL and above DSL, the novel feasibility constraint applied in RSDP-F and RSDP-S models is written as Eq. (11):

$$\max(R_{\min}, S_t + Q_t - S_{\max}) \leq R_t \leq \min(R_{\max}, S_t + Q_t - S_{\min,R}) \quad (11)$$

As previously mentioned, solution robustness often adopts higher-order moments of the objective value either from its best value or from its most frequent value. Among these two candidates, this study defined solution robustness as the deviation from the best possible value across all scenarios due to the fact that the most frequent value was already considered in the recursive equation of SDP [Eq. (3)] through the multiplication of transition probabilities. Consequently, the third term of the aggregate objective function, which corresponds to the key concept of solution robustness, measures the deviation of the main objective term of specific scenarios s ($f_{1,s,t}$) from the best possible main objective across all inflow scenarios S ($\min_{s \in S} f_{1,s,t}$)

$$f_3 = \left(\min_{s \in S} f_{1,s,t} - f_{1,s,t} \right) \quad (12)$$

Finally, the last term in the aggregate objective function is included in order to prevent the optimization algorithm from finding solutions that may lead to depletion of remaining storage even after the first objective is satisfied. In real-world simulations, the dam elevation at the end of the drawdown period, $S_{t,end}$, is required to meet a specific standard elevation for the upcoming flooding season, S_{target} , after releasing water for M&I uses. To enhance dominance on the first objective f_1 , the last objective f_4 is included with a relatively smaller weight ($w_1, w_2, w_3 = 10^6$, $w_4 = 1$) than the weights of other objective terms

$$f_4 = \frac{S_{t,end} - S_{target}}{S_{target}} \quad (13)$$

By aggregating the four objective terms with predefined weights in the models, $\mathbf{w} = (w_1, w_2, w_3, w_4)$ (Table 1), the objective function, $B_t(\cdot)$ for each model can be written using the following equation:

$$B_t(S_t, Q_t, R_t) = w_1 f_1^2 + w_2 f_2^2 + w_3 f_3^2 + w_4 f_4^2 \quad (14)$$

Calculation

Prior to iteratively calculating Eq. (3) with Eq. (14), the state variables of the SDP model, S_t and Q_t , were discretized to k and l intervals, respectively. In this study, the authors selected the discretization intervals in SDP models to be $k = 30$ and $l = 11$ so that they can sufficiently cover the whole range of variables densely. Consequently, the total number of paths of the problem is calculated by multiplying the discretization intervals for $S(t)$, $S(t+1)$, $Q(t)$, t , and n , where t and n are the number of months and iterative calculations of the recursive equation, respectively. In this study, the total number of paths to be calculated for each model depends

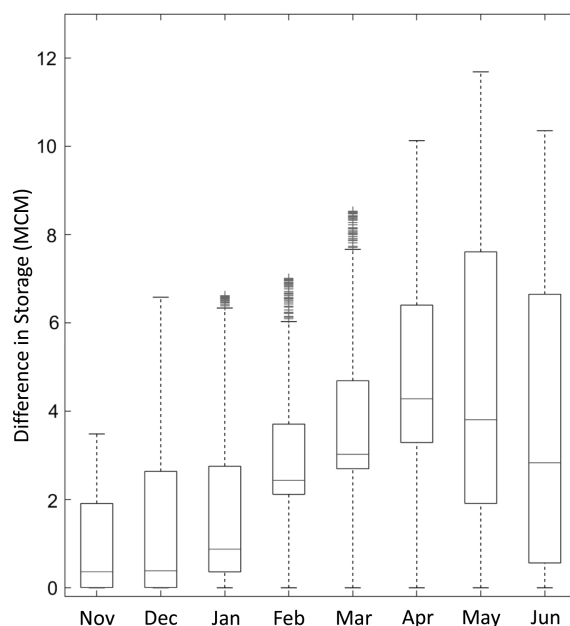
Table 2. Ratio of hedging according to the zone-based hedging rule applied to all multipurpose dams in Korea

Zone name	Municipal and industrial (%)	Irrigation (%)	Environmental (%)
Normal zone	100 ^a	100 ^a	100 ^a
Zone 1 (attention zone)	100	100	100
Zone 2 (caution zone)	100	100	Up to 0
Zone 3 (alert zone)	100	Up to 70	0
Zone 4 (critical zone)	Up to 80	70	0

^aRelease exceeding demand is possible in normal zones.

on the number of iterations for each model n_{model} : $k \times k \times l \times t \times n_{model} = 30 \times 30 \times 11 \times 9 \times n_{model}$. This is a relatively conservative number of calculation that could be conducted without parallelization, unless n_{model} increases exponentially.

The optimal policies derived from the three SDP models are coupled with the zone-based hedging rule, which is adopted as the main operating policy for all multipurpose dams in Korea, to derive policies that reflect how the actual reservoirs are operated. For comparative analysis, two optimal rules both derived by the CSDP model are simulated under a wide range of synthetic inflow series. While the first set of rules is derived directly according to the results of optimization, the second candidate rule set coupled the real-world zone-based hedging rule (Table 2) during all steps of optimization. Consequently, the second set yielded reduced release decisions especially in lower zones of storage (Zone 3 and 4 of Table 2). Then, both rules underwent 50,000 years of baseline simulation and the values of beginning-of-month storage were analyzed (Fig. 4). The difference between the storage values of the two sets suggest that the rule set that took hedging into consideration is more effective in terms of long-term response to drought since the release is regulated with respect to current storage. Since the difference between the two during March to May ($3.9 \times 10^6 \text{ m}^3$) is about half of the average monthly demand ($8.7 \times 10^6 \text{ m}^3$), taking hedging into consideration during optimization procedures would

**Fig. 4.** Boxplots of the difference in beginning-of-month storage after 50,000 years of simulation between optimized release rules and reduced release rules with hedging.

result in significant difference in storage after simulation. This study simplified, yet endeavored to find, a balanced operation rule between theoretically optimal solutions and real-world policies in the optimization step. Indeed, a more delicate and detailed approach regarding integration is possible and remains as a potential future research topic.

All release decisions R_t were derived according to optimization models that were coupled with hedging rules before entering the continuity equation for the reservoir, Eq. (15):

$$S_{t+1} = S_t + Q_t - R_t \quad (15)$$

Among 22 historic flooding seasons, the study reservoir only failed to reach the 90% of normal high water level (NHWL) only in 2015 (89%) and 2019 (86%) (i.e., 20 out of 22 years managed to reach the 90% of NHWL during their flooding seasons). Therefore, it was assumed that at the beginning of the drawdown period, the reservoir had been supplied with enough water from flooding and start the drawdown period from NHWL. Thus, only the rules for the 9-month drawdown period were considered, mainly due to the unique monsoonal climate of East Asia where precipitation is concentrated during the 3-month flooding season (Kim et al. 2007). The final outputs were tables of steady-state release rules, which are interpretable by stakeholders without expertise. These rules satisfy the following equation:

$$\begin{aligned} & \|f_{opt}(S_t^k, H_t)_{n_{model}} - f_{opt}(S_t^k, H_t)_{n_{model}-1}\| \\ & - \|f_{opt}(S_t^k, H_t)_{n_{model}-1} - f_{opt}(S_t^k, H_t)_{n_{model}-2}\| < 10^{-5} \quad \forall t, k \end{aligned} \quad (16)$$

Simulation

Inflow Series Generation

Recently, in the field of water systems operations, scenario-led approaches, in which the system performance is evaluated using global climate projection scenarios (Wilby and Dessai 2010), are being replaced by scenario-neutral approaches, in which the assessment is conducted under a wide range of uncertain scenarios sampled from deep uncertainty (Kwakkel et al. 2016; Biglarbeigi et al. 2018). In accordance, this study generated a wider range of inflow scenarios so that the envelope of the simulation sufficiently covers the scope of the projections.

Optimized release rules were tested under future inflow scenarios that capture the nonstationarity of climate change. Second-order nonstationarity was assumed in future inflow scenarios; consequently, this study generated future inflow series by only modifying the mean and standard deviations of the historic inflow series. In order to select the range of parameters, the statistics of the most recently available 27 general circulation model (GCM) streamflow projections in which the study area is located were computed (Seo and Kim 2018). When compared with the statistics from historical observations, the statistics of the GCM scenarios fell within a range of 0.5–1.5 times of the observations (Fig. 5). Consequently, in this study, the authors expanded this range, so that the generated inflow scenarios would cover a wider range between 0.1 and 2.0, in order to account for possible extremes that may occur under climate change (Salman et al. 2018; Farjad et al. 2019).

Specifically, for inflow series generation, this study first assumed that the historical monthly inflow series of the study basin, Q_i , follow a lognormal distribution and fitted each month's inflow to a lognormal distribution. Then, in order to synthetically generate inflow series X_i with varying mean and standard deviations, the

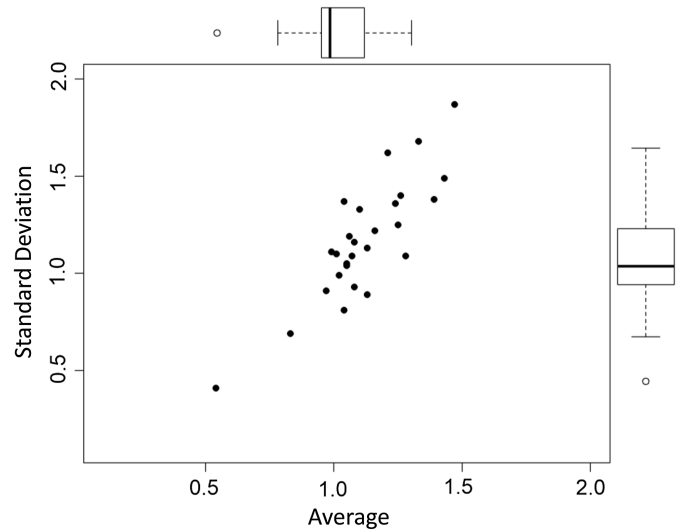


Fig. 5. Range of average and standard deviations of selected GCM scenarios. This study generated a total of 400 cases of synthetic inflow series in which the average and standard deviations are varied with corresponding ratios (k_μ and k_σ), ranging from 0.1 to 2.0, so that the synthetic inflow would cover the entire range of GCM scenarios.

parameters of the log transformed synthetic inflows ($Y_i = \ln(X_i + \epsilon)$), $\bar{Y}_i$ and s_{Y_i} were estimated using Eqs. (17) and (18) (Yoon et al. 2018)

$$s_{Y_i} = \ln[1 + k_\sigma \times s_{Q_i} (k_\mu \times \bar{Q}_i + \epsilon)^2] \quad (17)$$

$$\bar{Y}_i = \ln(k_\mu \times \bar{Q}_i + \epsilon) - 12s_{Y_i}^2 \quad (18)$$

where i is the index of the month and $\epsilon = 0.001$ is a small number added during log transformation, $\bar{Y}_i$ and s_{Y_i} are the mean and standard deviations of the synthetically generated inflow, $\bar{Q}_i$ and s_{Q_i} are the mean and standard deviation of the historical inflow, and k_μ and k_σ are the multipliers ranging from 0.1 to 2.0 each multiplied by the original parameters for synthetic generation.

Also, in order to maintain the lag-1 correlation between monthly inflow scenarios, the covariance matrix of the log transformed inflow between consecutive months i and $i+1$, $\hat{\Sigma}_{Y_i, Y_{i+1}}$ was estimated using Eq. (19)

$$\hat{\Sigma}_{Y_i, Y_{i+1}} = \hat{\rho}_{\ln Q_i, \ln Q_{i+1}} \times s_{Y_i} \times s_{Y_{i+1}} \quad (19)$$

where $\hat{\rho}_{\ln Q_i, \ln Q_{i+1}}$ is the correlation matrix between the log transformed historical inflow of month i and $i+1$.

As the synthetic log transformed inflow Y_i follows the multivariate normal distribution [$Y_i \sim MVN(\bar{Y}_i, \hat{\Sigma}_{Y_i, Y_{i+1}})$], this study adopted the mvnrnd function in MATLAB and created 50,000 continuous yearly inflow scenarios for each statistical scenario in which the average and standard deviation of the scenarios varied from those of historical observations. Finally, to compare all models fairly and to minimize the effects of outliers on the evaluation metrics, identical sets of inflow scenarios were used during the simulation of all models.

Reservoir Operation Simulation

For simulation, each optimization model underwent 50,000 years of simulation for each inflow scenario, in which the optimized release rules were applied. The main governing equation adopted during the simulation process is Eq. (15), and monthly averaged

Table 3. Selected traditional evaluation metrics in this study

Metric name	Classification	Equation	Unit
ADM_y	Magnitude of annual water deficit	$\frac{1}{T} \sum_{t=1}^T \max\left(\frac{D_t - R_t}{D_t}, 0\right)$	Unit-less
ADF_y	Frequency of annual water deficit	$\frac{1}{T} \sum_{t=1}^T I_t\left(\frac{D_t - R_t}{D_t} < 0\right)$	Unit-less
$ReT_{\max}$	Duration of longest failure	$\begin{aligned} &\text{Max}_t \{t_b - t_a\}, \quad \forall t_a \\ &\text{s.t. } \frac{D_{t_a} - R_{t_a}}{D_{t_a}} > 0 \text{ and } \frac{D_{t_a-1} - R_{t_a-1}}{D_{t_a-1}} \leq 0 \\ &\text{and for first } t_b > t_a \\ &\frac{D_{t_b} - R_{t_b}}{D_{t_b}} > 0 \text{ and } \frac{D_{t_b-1} - R_{t_b-1}}{D_{t_b-1}} \leq 0 \end{aligned}$	Months

Note: ADM = average deficit magnitude; ADF = average deficit frequency; ReT = recovery time; and $I_t(\cdot)$ = counts the number of occurrences inside the parentheses.

performance metrics were calculated for further analysis of simulation results.

Evaluation

After the simulation, the evaluation process of this paper was completed in two steps. Step 1 focused on using traditional metrics originally proposed by Hashimoto et al. (1982), and Step 2 focused on using metrics of the family of robustness metrics, in which a system's performance under deeply uncertain or less probable states of the world can be sufficiently evaluated (Maier et al. 2016). The primary motivation behind presenting the results using both traditional and robustness metrics is to possibly reduce human decision biases, such as cognitive myopia or short-sightedness (Hogarth 1981). This could also be used to broaden the understanding of the policies' operational performance and their ability to perform acceptably under challenging system states.

Traditional Metrics

In this paper, the concepts of reliability, resiliency, and vulnerability originally suggested by Hashimoto et al. (1982) were adopted to measure the frequency, duration, and magnitude of the failure of the system, respectively. These three concepts were widely adopted to evaluate the performances of many engineering systems, and also provided the basis for variants of metrics. For instance, Onta et al. (1991) adopted the concept of reliability from Hashimoto et al. (1982) for the assessment of alternative plans and policies for conjunctive use of surface and groundwater resources. Srdjevic et al. (2004) applied this methodology to evaluate simulations of multi-year water management scenarios, and El-Baroudy and Simonovic (2004) proposed a coupled measure that combined classic reliability and the fuzzy concept and used it to evaluate the performance of water resource systems.

To distinguish from another family of metrics to be introduced, this paper will refer to the reliability, resiliency, and vulnerability metrics as "traditional" metrics. As the main objective of the target reservoir is to supply water for the general public for M&I uses, this paper defined failure as all incidents in which the actual release, R_t , was not able to meet the target demand, D_t , of the specific month t (i.e., $R_t < D_t$).

For intuitive and consistent demonstration, all three metrics that were adopted in this study were devised based on Hashimoto et al. (1982) with slight modification. Average deficit magnitude (ADM_y), average deficit frequency (ADF_y), and maximum recovery time ($ReT_{\max}$) correspond to vulnerability, reliability, and

resiliency from Hashimoto et al. (1982), respectively. All metrics are annually averaged values and also standardized so that all metrics with higher values indicate events with greater, more frequent, and longer water deficits. ADM_y represents the ratio of annual average water deficit magnitudes measured as a fraction of the total deficit over total demand; ADF_y indicates the annual average frequency of the water deficit, measured as a fraction of the number of failures over the total time step; and $ReT_{\max}$ is the longest time step of each scenario for the system to come back from the state of failure to normal, measured in months. Mathematical formulas for the three traditional metrics adopted in this study are shown in Table 3.

Robustness Metrics

Although RSDP-F and RSDP-S models were originally designed so that they would yield relatively more robust solutions than the CSDP model, this superiority may not be necessarily prominent when a system's performance is solely measured using traditional metrics. Moreover, owing to the complexity of most water resources system problems, the first procedure in the evaluation of robustness starts with the meta-problem of deciding how to decide (Schneller and Sphicas 1983; Reed and Kasprzyk 2009). As the principal motivation of this paper was not in determining sophisticated robustness measures, but in the development of both optimal and robust operating policies and testing them under a wide range of future uncertainties, this paper adopted the existing concept of robustness in the field of decision making for water resources for its evaluation.

A myriad of robustness metrics in this specific field of study can be classified into different categories depending on the selected measures. Especially when the extent of risk aversion subsumed into the mathematical formula is considered, the metrics can be largely divided into either regret-based (Savage 1951) metrics or satisficing-based (Simon 1956) metrics. Regret-based metrics, in this specific context, measure the relative difference between the selected alternative performance and that of the best (or most probable) alternative inside the envelope of simulation (McPhail et al. 2018). In contrast, satisficing metrics evaluate the tendency of decision makers to select alternatives that meet their requirements instead of being optimal according to other criteria (Giuliani and Castelletti 2016).

Thus, in this paper, with a focus on classical risk measures and risk aversion, a regret-based robustness measure was analyzed. The regret metric defined in this study, R^i , measures the 90th percentile deviation from the average performance using a baseline scenario

Table 4. Number of possible paths in each optimization model

Model	i	j	t	k	rep_n	Number of paths = $i \times j \times t \times k \times rep_n$
CSDP	30	30	9	11	62	5,524,200
RSDP-F	30	30	9	11	65	5,791,500
RSDP-S	30	30	9	11	68	6,058,800

(Herman et al. 2015) with respect to each traditional metric D_i ($i = 1, 2, 3$). As regret metrics are separately calculated with respect to reliability, resiliency, and vulnerability, a total of three regret-based robustness measures for each model can be computed. The mathematical formula of the robustness metric primarily used in this study, R_{model}^i , is derived using Eqs. (20) and (21)

$$R_{model}^i = [D_{i,90} : P(D_i \leq D_{i,90}) = 0.90] \quad (20)$$

$$D_{i,j} = \frac{|F(\bar{x})_{i,j} - F(\bar{x})_i^*|}{F(\bar{x})_i^*} \quad (21)$$

where $F(\bar{x})_i^*$ is the annual averaged value of metric i of the baseline state of the world; and $F(\bar{x})_{i,j}$ is the annual averaged value of metric i calculated for the j th state of the world.

Results

Increase in Number of Feasible Paths

As previously mentioned, the first advantage of the introduction of soft constraints with RSDP models is in the expansion of the range of potential solutions to be considered. After calculation and derivation of optimal monthly operating policies, the total number of calculations, or possible paths of SDP, for each model can be summarized as shown in Table 4.

For the CSDP model, 96.67% of the total number of paths passed through the feasibility constraint given in Eq. (2) and were considered as candidates for the optimal solution. Similarly, 96.67% of the state variables of RSDP-F and RSDP-S models passed through the feasibility constraint of RSDP [Eq. (11)], and when the feasibility was checked by applying the CSDP standards, 93.33% of all state variables passed the former constraint. This indicates that the differences between the two, 3.34% of all state variables in RSDP models, which were originally neglected when optimized under the CSDP feasibility constraint, are now considered as potential solutions in RSDP models. Although an enhancement of the simulation results measured by either traditional or robustness metrics was the primary motivation behind the introduction of the RSDP models, boosting in the range of possible solutions would potentially increase the number of candidate solutions. This enlargement in the range of solutions, regardless of their optimality, would ultimately allow both the simulation tool and the related decision-makers more flexibility under various circumstances.

Sample Operation under Specific Drought Incidents

The optimized release rules of CSDP, RSDP-F, and RSDP-S models were simulated under two sets of 10-year drought incidents both extracted from the baseline scenario. Historically during the 2014–2019 drought, the study reservoir had to undergo inflow conditions that correspond to the 87th percentile's worst case among 50,000 years of the baseline scenario. Thus, the first set of representative 10-year drought assumed the historical recorded drought conditions to last for 10 years, while the second set assumed the worst-case drought conditions extracted from the 50,000 years of the baseline scenario.

The release rules and the main objective of the three models were tested in order to compare their distinctive behaviors.

Release decisions of the three models varied the most when the storage level was low. While the RSDP-S model yielded rules with significantly reduced release under dry conditions, CSDP and RSDP-F models lowered releases, yet not so significantly when compared with the decisions made under RSDP-S model. This difference consequently led to greater magnitude in deficit near the end of drawdown periods where the demand was the highest among other months. When compared in terms of cumulative water deficit (f_1) (Fig. 6), operating the reservoir under the RSDP-S model's policy would potentially lead to better performance in the long term.

Evaluation through Traditional Metrics

The analysis of the system via traditional metrics aims to visualize the effects of adopting either the RSDP-F or RSDP-S model versus the CSDP model in terms of magnitude (ADM_y), frequency (ADF_y), and duration (ReT_{max}) of water supply deficit across the entire envelope of the simulation range. As traditional metrics or similar variations have been widely adopted in decision making in real-world systems, a relatively intuitive display of the three selected metrics was visualized through contour plots for each model (Fig. 7). Although the range of ratios of both parameters (k_μ and k_σ) was from 0.1 to 2.0, only the results under scenarios with decreasing average ($k_\mu = 0.1$ – 1.0) are displayed mainly to reflect the severe multiyear drought the reservoir underwent recently and its consequences due to high ratio of M&I water use. In all plots in Fig. 7, the change in colors in the horizontal directions indicates that the system is more sensitive to a change in k_μ than that of k_σ 's, except for specific cases where k_μ is extremely high or extremely low.

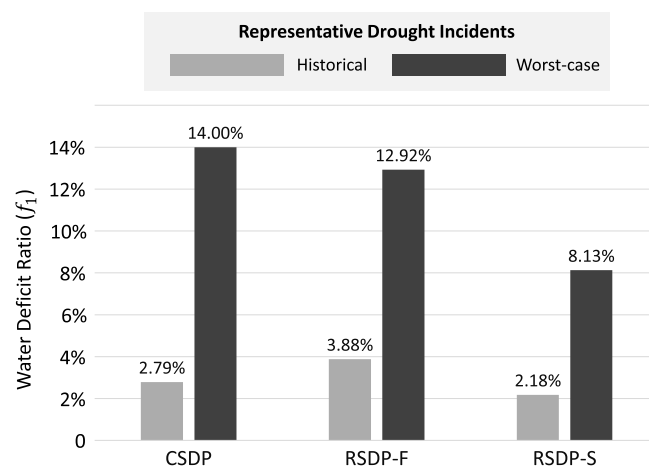


Fig. 6. Water deficit ratio (f_1) of three models over 10 years of representative drought incidents extracted from the baseline scenario: historical (2014–2019) and worst-case.

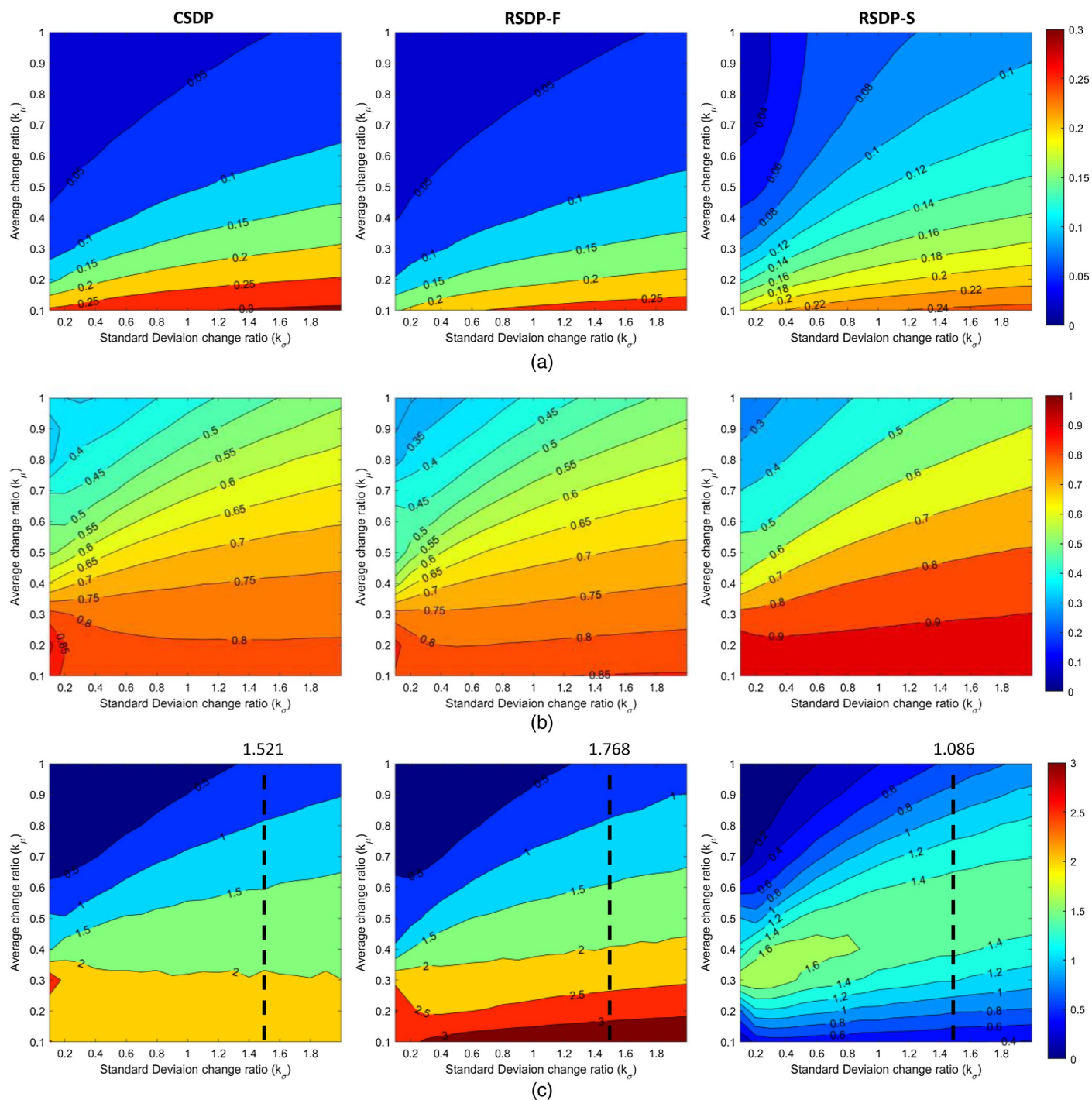


Fig. 7. Contour plots of traditional metrics across synthetic inflow scenarios with scenarios with decreasing average: (a) ADM_y ; (b) ADF_y ; and (c) ReT_{max} .

Both RSDP-F and RSDP-S models outperformed the CSDP model in different metrics, and also in different circumstances. Especially in extreme inflow cases with a decreased average and an increased standard deviation (bottom-right corner of Fig. 7), both models outperformed the CSDP model in terms of magnitude [Fig. 7(a)], and the RSDP-S model outperformed the CSDP model in terms of the duration of longest failure [Fig. 7(c)]. A significant trade-off relationship between frequency [Fig. 7(b)] and the other two metrics also can be observed, where the RSDP-S model performed slightly worse than the other two models. Although the CSDP model appears to be the optimal policy in terms of magnitude, as the

main objective term of the CSDP model is to minimize the magnitude of water deficit, it was proven that the system may undergo failures of greater magnitude in extreme cases.

To demonstrate how the elicitation of robust alternatives would affect the related stakeholders, this study addressed the problem in two ways. The first approach aims to analyze the values of traditional metrics to answer the following question: how extreme are the extremes? The second idea includes the use of robustness metrics and will be presented in the following subsection. For instance, in terms of the longest duration, the RSDP-S model outperformed both the CSDP and RSDP-F models when measured with the mean

value across all scenarios with $k_{\sigma} = 1.5$ [dashed lines in Fig. 7(c)]. The difference in the $\text{Re}T_{\max}$ value of the RSDP-S and CSDP models, 0.435 (1.521–1.086), indicates that the longest duration for water deficit may differ by more than 13 days. As the primary water use from this reservoir is for M&I use, failing to provide water for 13 days may result in critical consequences in real systems. Thus, despite the complexity and potential cost that may occur in transforming real-world policies, in which robustness is not yet considered, shifting towards robust problem formulations appears to be necessary and may lead to better outcomes when measured with traditional metrics.

How Robust Are Robust Policies?

A closer examination of the simulation results was made possible through the use of regret-based robustness metrics, R_i^{model} , which measure the relative difference between the performance in the baseline scenario and the performance under the 90th percentile scenario. The parallel coordinates plot in Fig. 8 illustrates the trade-off relationships between different objectives or metrics according to different policies.

In Fig. 8, each vertical axis on the plot reports on the robustness of each model in terms of magnitude (R^{Mag}), frequency (R^{Freq}), and duration (R^{Dur}). An ideally robust policy, consequently, would be a straight line at the bottom of the plot, as displayed in the gray line with circles in Fig. 8. When compared with the models, the CSDP model and RSDP-F model appear to perform similarly in terms of R^{Freq} and R^{Dur} , whereas RSDP-F outperforms CSDP in terms of R^{Mag} . This advantage of the RSDP-F model mainly comes from the inclusion of feasibility robustness into the objective function and the constraint. In contrast, in the RSDP-S model, in which the magnitude of the variance in the main objective is penalized, robustness metrics measured in magnitude and duration unquestionably surpassed those of the other two models. The difference in R^{Dur} indicates that when compared with the baseline scenario, operating with CSDP policies may lead to two times longer extremes than the results of operating using the RSDP-S rules.

Although the simulation results evaluated via traditional metrics may seem explicit to determine which model performed better, the robustness metrics lead to the conclusion that operation under

RSDP-S policies would yield far more robust outcomes than operating with conventional policies that do not include any concept of robustness into the problem formulation processes. Nevertheless, as the crossings in lines displayed in Fig. 8 indicate a trade-off relationship between the three robustness metrics, the selection of models according to stakeholder preference is also possible. From another perspective, the results of the evaluation using the robustness metric also indicate that the current conventional policies can be extremely vulnerable to potential extremes, again highlighting the necessity of shifting away from conventional decision-making strategies.

Conclusion

This study adds RO into the problem formulation process of deriving optimal and robust reservoir operation policies to confront the nonstationarity of climate change. Due to SDP's popularity and wide applicability in real-world reservoir operation problems, key concepts of RO were incorporated into both the objective function and the constraints of SDP. The proposed methodology was applied to deriving release policies for a multipurpose Boryeong Dam located in Korea, of which the main objective is to supply M&I water to nearby cities and districts. Through relating the theoretical concepts with real-world policies, this study was able to derive three sets of policies for comparative analysis. The simulation of conventional and robust policies was conducted under a wide range of climate conditions, and the results indicate that RSDP policies, especially the RSDP-S model in which the solution robustness is incorporated into the model formulation, outperform CSDP policies in both traditional and robustness metrics in extreme climate conditions. Specifically, electing RSDP-S model would result in half the duration of water deficit than that of the CSDP model in terms of regret-based robustness metrics. Thus, the results of this study once again highlight the fundamental concept of decision-making strategies under climate change: a separate set of strategies should be contemplated and adopted after the analysis of their strengths and weaknesses (Watkins and Hunt 2013). Nevertheless, since the current analysis only reflects a constrained range of future with decreased inflow, the approach in this study could be further expanded to better reflect the projected future GCM scenarios where the mean inflow increases.

This study was the first attempt in which the concept of RO proposed by Mulvey et al. (1995) was linked to SDP in a reservoir operation problem. Also, through the inclusion of real-world discrete hedging policies into both optimization and simulation procedures, this study was able to suggest sets of candidate policies that are applicable to real-world reservoirs. Nevertheless, multiple potential extensions of this research are possible, such as connections to finance or authority sides of the system, especially shadow prices in water finance (Molinos-Senante et al. 2010; Bierkens et al. 2019) or power asymmetries triggered by different interest groups (Denaro et al. 2018), which can be closely related to the proposed methodology and case study. Regarding the algorithmic implications of considering broader sources of problem complexity (e.g., additional state variables, exogenous information feedbacks, large numbers of conflicting objectives, stochastic simulation, etc.), more recent multi-objective policy-based control strategies hold promise (Giuliani et al. 2016; Quinn et al. 2019).

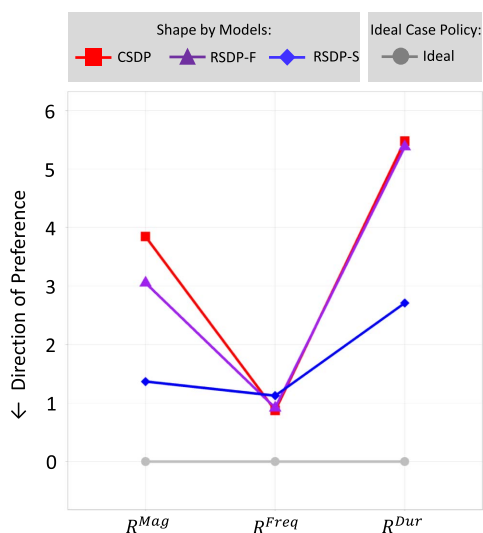


Fig. 8. Parallel coordinates plot of robustness metrics by model. R^{Mag} is the robustness metric measured in terms of magnitude, R^{Freq} is the robustness metric measured in terms of frequency, and R^{Dur} is the metric measured in terms of duration.

Data Availability Statement

All the data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request.

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